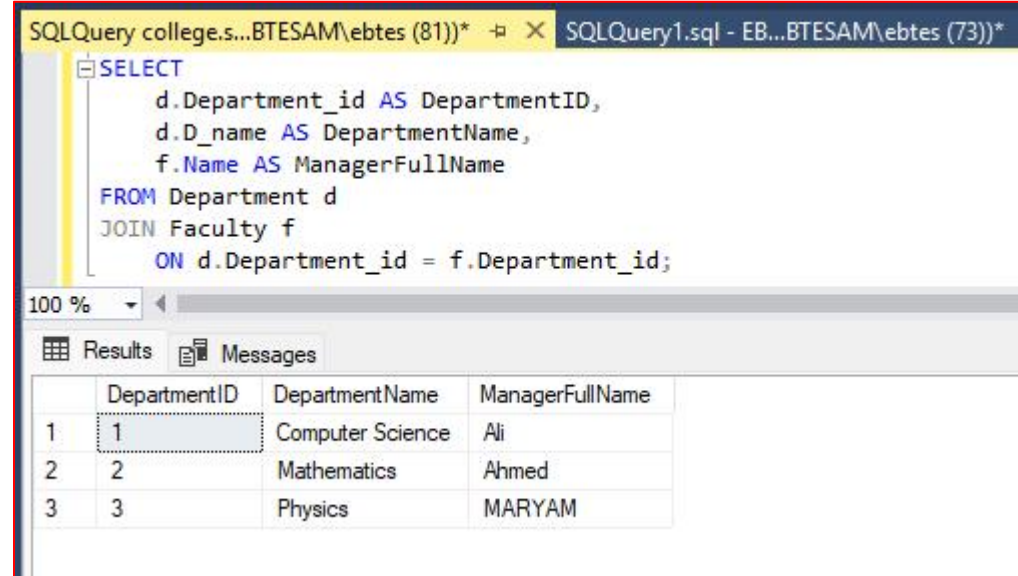


University Database – JOIN Queries

1. Display the department ID, name, and the full name of the faculty managing it.



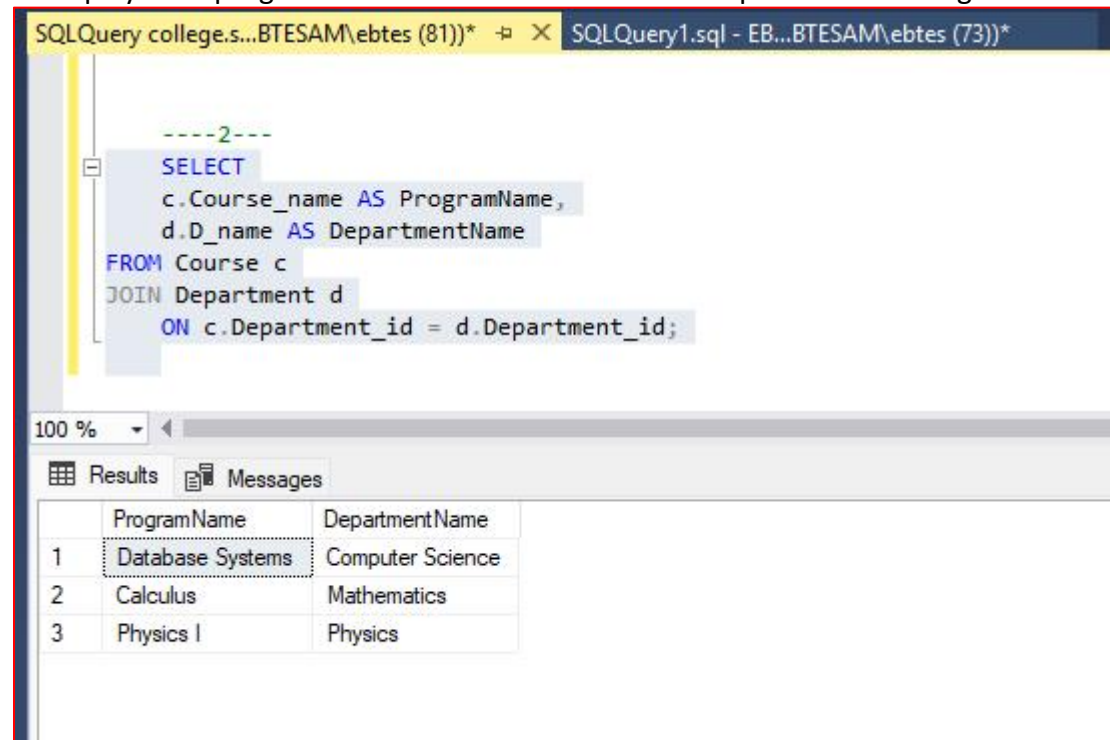
The screenshot shows a SQL query window with the following query:

```
SELECT
    d.Department_id AS DepartmentID,
    d.D_name AS DepartmentName,
    f.Name AS ManagerFullName
FROM Department d
JOIN Faculty f
    ON d.Department_id = f.Department_id;
```

The results are displayed in a table with the following data:

	DepartmentID	DepartmentName	ManagerFullName
1	1	Computer Science	Ali
2	2	Mathematics	Ahmed
3	3	Physics	MARYAM

2. Display each program's name and the name of the department offering it.



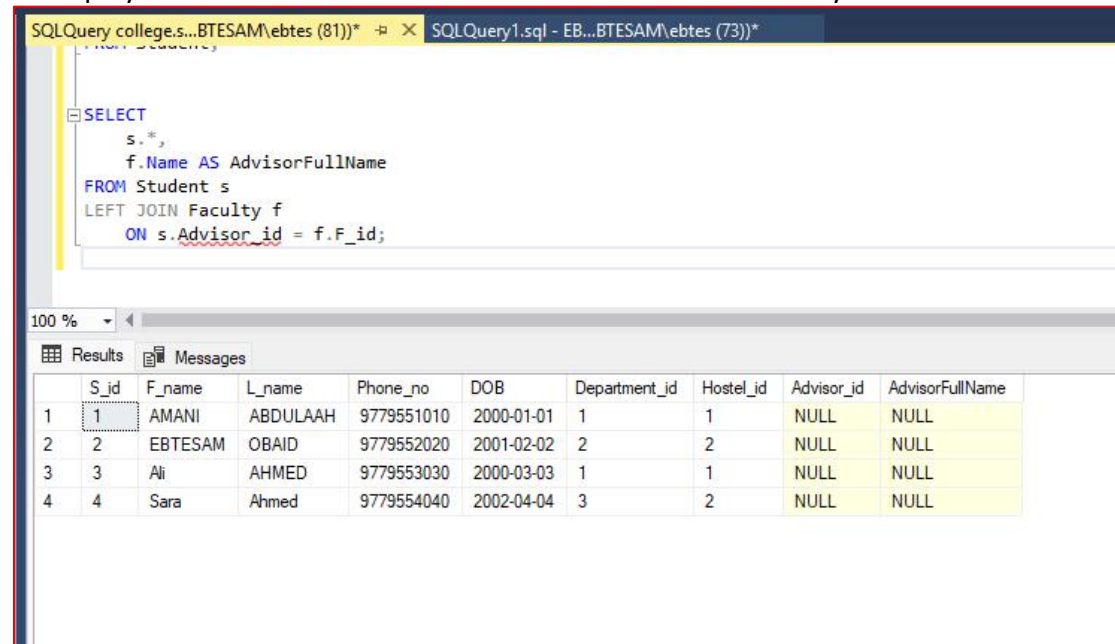
The screenshot shows a SQL query window with the following query:

```
-----2-----
SELECT
    c.Course_name AS ProgramName,
    d.D_name AS DepartmentName
FROM Course c
JOIN Department d
    ON c.Department_id = d.Department_id;
```

The results are displayed in a table with the following data:

	ProgramName	DepartmentName
1	Database Systems	Computer Science
2	Calculus	Mathematics
3	Physics I	Physics

3. Display the full student data and the full name of their faculty advisor.



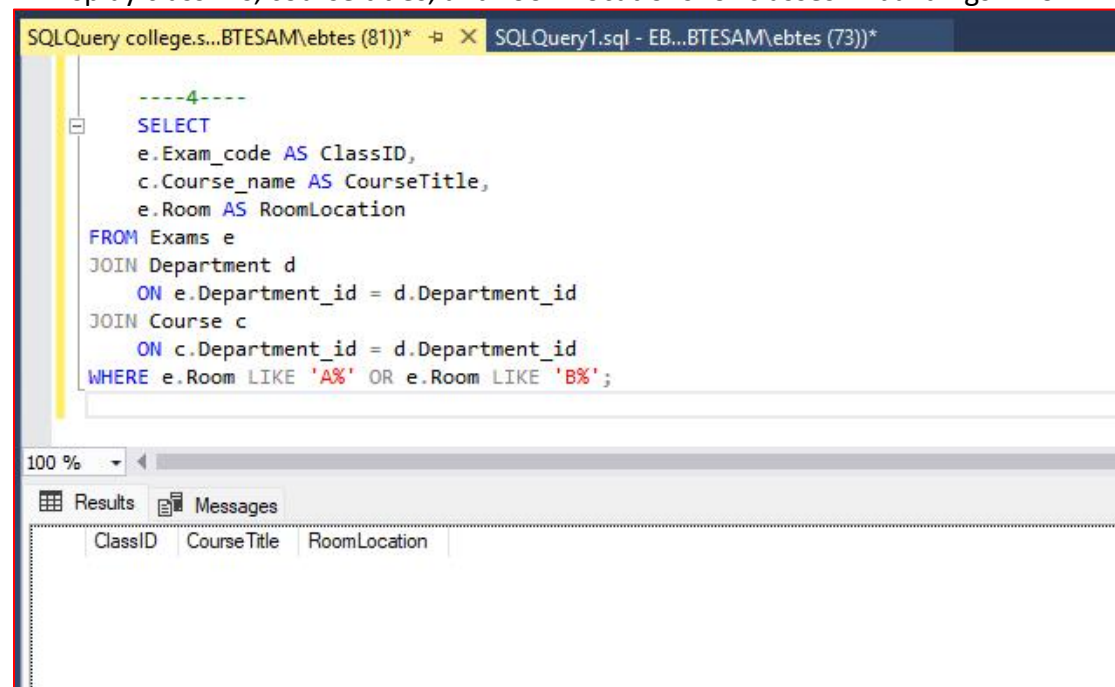
The screenshot shows a SQL query window with the following query:

```
SELECT
    s.*,
    f.Name AS AdvisorFullName
FROM Student s
LEFT JOIN Faculty f
    ON s.Advisor_id = f.F_id;
```

The results pane displays a table with the following data:

	S_id	F_name	L_name	Phone_no	DOB	Department_id	Hostel_id	Advisor_id	AdvisorFullName
1	1	AMANI	ABDULAAH	9779551010	2000-01-01	1	1	NULL	NULL
2	2	EBTESAM	OBAID	9779552020	2001-02-02	2	2	NULL	NULL
3	3	Ali	AHMED	9779553030	2000-03-03	1	1	NULL	NULL
4	4	Sara	Ahmed	9779554040	2002-04-04	3	2	NULL	NULL

4. Display class IDs, course titles, and room locations for classes in buildings 'A' or 'B'.



The screenshot shows a SQL query window with the following query:

```
SELECT
    e.Exam_code AS ClassID,
    c.Course_name AS CourseTitle,
    e.Room AS RoomLocation
FROM Exams e
JOIN Department d
    ON e.Department_id = d.Department_id
JOIN Course c
    ON c.Department_id = d.Department_id
WHERE e.Room LIKE 'A%' OR e.Room LIKE 'B%';
```

The results pane displays a table with the following data:

ClassID	CourseTitle	RoomLocation
---------	-------------	--------------

5. Display full data about courses whose titles start with "I" (e.g., "Introduction to...").

SQLQuery college.s...BTESAM\ebtes (81))* SQLQuery1.sql - EB...BTESAM\ebtes (73))*

```
--5-----
SELECT *
FROM Course
WHERE Course_name LIKE 'I%';
```

100 %

Results Messages

	Course_id	Course_name	Duration	Department_id
1	1	Database Systems	4 months	1

6. Display names of students in program ID 3 whose GPA is between 2.5 and 3.5.

SQLQuery college.s...BTESAM\ebtes (81))* SQLQuery1.sql - EB...BTESAM\ebtes (73))*

```
SELECT TOP 5 *
FROM Student;

UPDATE Student
SET ProgramID = 3, GPA = 3.0
WHERE S_id IN (/* list of student IDs */);

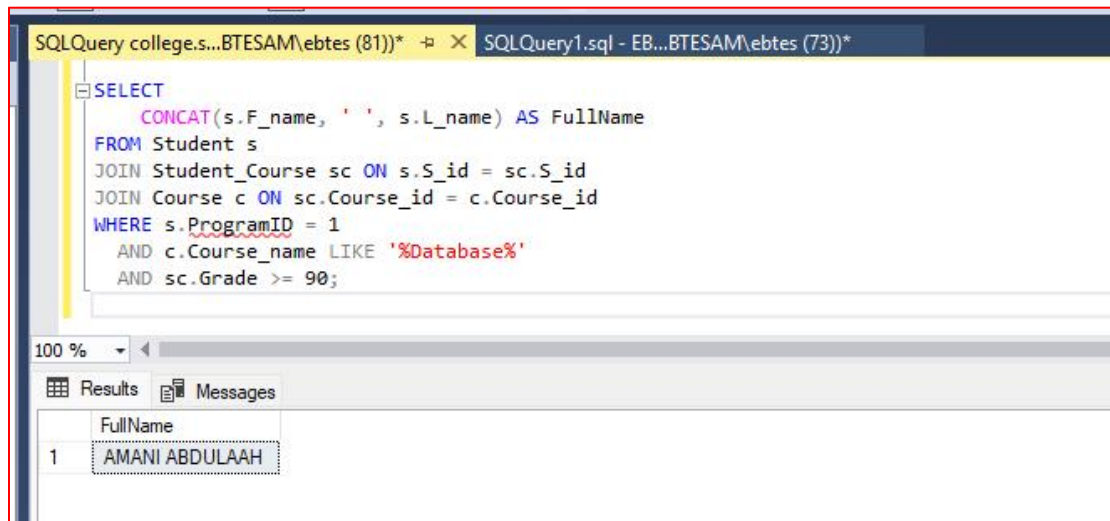
SELECT
    CONCAT(F_name, ' ', L_name) AS FullName
FROM Student
WHERE ProgramID = 3
AND GPA BETWEEN 2.5 AND 3.5;
```

100 %

Results Messages

	S_id	F_name	L_name	Phone_no	DOB	Department_id	Hostel_id	EnrollmentDate	CurrentStatus	GPA	Advisor_id
1	1	AMANI	ABDULAAH	9779551010	2000-01-01	1	1	2020-09-01	Active	3.50	NULL
2	2	EBTESAM	OBAID	9779552020	2001-02-02	2	2	2021-01-15	Active	3.75	NULL
3	3	Ali	AHMED	9779553030	2000-03-03	1	1	2022-05-10	Active	3.20	NULL
4	4	Sara	Ahmed	9779554040	2002-04-04	3	2	2023-08-20	Active	3.90	NULL
5	300046	YourFriendFirst	YourFriendLast	NULL	NULL	NULL	NULL	NULL	NULL	2.00	NULL

7. Retrieve student names in the Engineering program who earned grades ≥ 90 in the "Database" course.



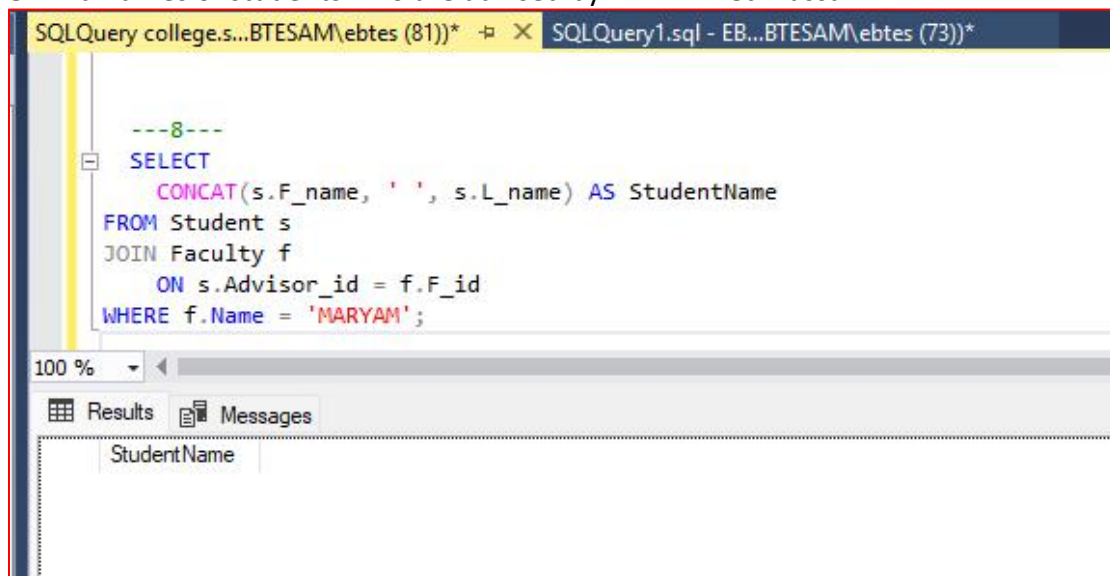
The screenshot shows a SQL query window with the following text:

```
SQLQuery college.s...BTESAM\ebtes (81))* SQLQuery1.sql - EB...BTESAM\ebtes (73))*  
  
SELECT  
    CONCAT(s.F_name, ' ', s.L_name) AS FullName  
FROM Student s  
JOIN Student_Course sc ON s.S_id = sc.S_id  
JOIN Course c ON sc.Course_id = c.Course_id  
WHERE s.ProgramID = 1  
    AND c.Course_name LIKE '%Database%'  
    AND sc.Grade >= 90;
```

Below the query, the 'Results' tab is active, displaying a single row of data:

FullName
1 AMANI ABDULAAH

8. Find names of students who are advised by "Dr. Ahmed Hassan".



The screenshot shows a SQL query window with the following text:

```
SQLQuery college.s...BTESAM\ebtes (81))* SQLQuery1.sql - EB...BTESAM\ebtes (73))*  
  
---8---  
SELECT  
    CONCAT(s.F_name, ' ', s.L_name) AS StudentName  
FROM Student s  
JOIN Faculty f  
    ON s.Advisor_id = f.F_id  
WHERE f.Name = 'MARYAM';
```

Below the query, the 'Results' tab is active, displaying a single column header:

Student Name

9. Retrieve each student's name and the titles of courses they are enrolled in, ordered by course title.

SQLQuery college.s...BTESAM\ebtes (81))* X SQLQuery1.sql - EB...BTESAM\ebtes (73))*

```

---9---
SELECT
    CONCAT(s.F_name, ' ', s.L_name) AS StudentName,
    c.Course_name AS CourseTitle
FROM Student s
JOIN Student_Course sc
    ON s.S_id = sc.S_id
JOIN Course c
    ON sc.Course_id = c.Course_id
ORDER BY c.Course name:

```

100 %

Results Messages

	StudentName	CourseTitle
1	EBTESAM OBAID	Calculus
2	AMANI ABDULAAH	Database Systems
3	Ali AHMED	Database Systems
4	Sara Ahmed	Physics I

10. For each class in Building 'Main', retrieve class ID, course name, department name, and faculty name teaching the class.

SQLQuery college.s...BTESAM\ebtes (81))* X SQLQuery1.sql - EB...BTESAM\ebtes (73))*

```

---10---
SELECT
    e.Exam_code AS ClassID,
    c.Course_name AS CourseName,
    d.D_name AS DepartmentName,
    f.Name AS FacultyName
FROM Exams e
JOIN Department d
    ON e.Department_id = d.Department_id
JOIN Course c
    ON c.Department_id = d.Department_id
JOIN Subject s
    ON s.Course_id = c.Course_id
JOIN Faculty f
    ON s.Faculty_id = f.F_id

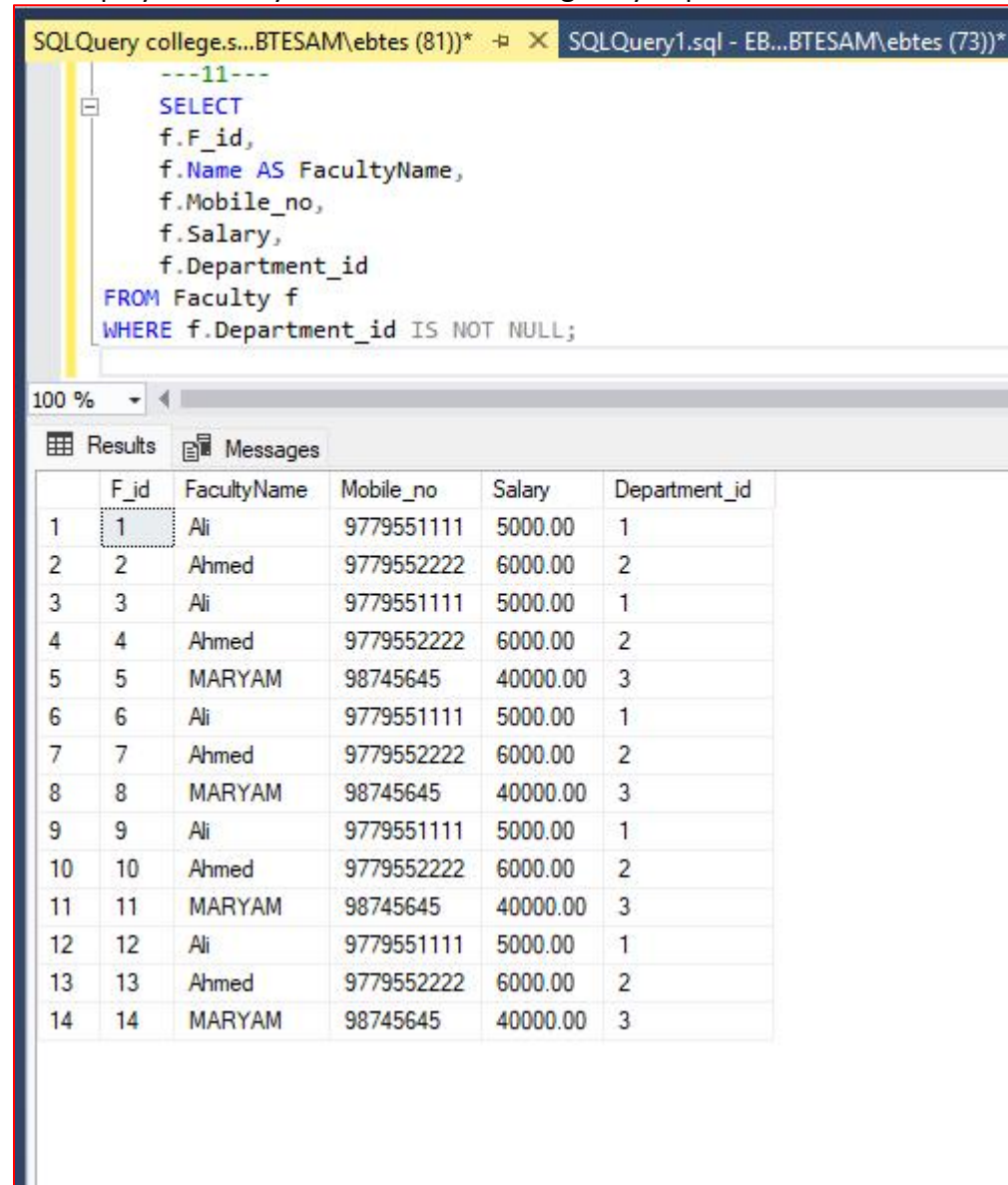
```

100 %

Results Messages

	ClassID	CourseName	DepartmentName	FacultyName
1	1	Database Systems	Computer Science	Ali
2	2	Calculus	Mathematics	Ahmed

11. Display all faculty members who manage any department.



The screenshot shows a SQL Server Enterprise Manager window with a query editor and a results pane. The query editor displays a SQL query to select faculty members who manage any department. The results pane shows a table with 14 rows of data.

SQLQuery college.s...BTESAM\ebtes (81))* X SQLQuery1.sql - EB...BTESAM\ebtes (73))*

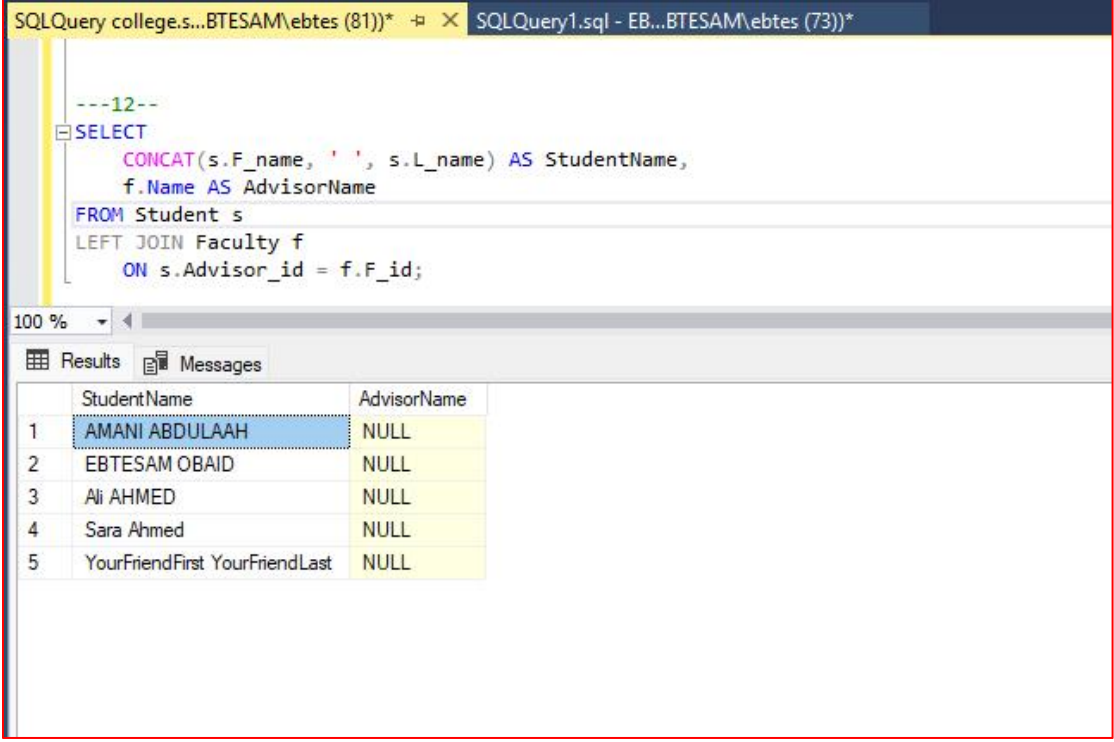
```
---11---  
SELECT  
f.F_id,  
f.Name AS FacultyName,  
f.Mobile_no,  
f.Salary,  
f.Department_id  
FROM Faculty f  
WHERE f.Department_id IS NOT NULL;
```

100 %

Results Messages

	F_id	FacultyName	Mobile_no	Salary	Department_id
1	1	Ali	9779551111	5000.00	1
2	2	Ahmed	9779552222	6000.00	2
3	3	Ali	9779551111	5000.00	1
4	4	Ahmed	9779552222	6000.00	2
5	5	MARYAM	98745645	40000.00	3
6	6	Ali	9779551111	5000.00	1
7	7	Ahmed	9779552222	6000.00	2
8	8	MARYAM	98745645	40000.00	3
9	9	Ali	9779551111	5000.00	1
10	10	Ahmed	9779552222	6000.00	2
11	11	MARYAM	98745645	40000.00	3
12	12	Ali	9779551111	5000.00	1
13	13	Ahmed	9779552222	6000.00	2
14	14	MARYAM	98745645	40000.00	3

12. Display all students and their advisors' names, even if some students don't have advisors yet.



The screenshot shows a SQL Server Enterprise Manager window with a query editor and a results pane. The query editor contains the following SQL code:

```
---12---  
SELECT  
    CONCAT(s.F_name, ' ', s.L_name) AS StudentName,  
    f.Name AS AdvisorName  
FROM Student s  
LEFT JOIN Faculty f  
    ON s.Advisor_id = f.F_id;
```

The results pane displays the output of the query, showing a table with two columns: StudentName and AdvisorName. The table contains five rows of data, with the first row highlighted in blue.

	StudentName	AdvisorName
1	AMANI ABDULAAH	NULL
2	EBTESAM OBAID	NULL
3	Ali AHMED	NULL
4	Sara Ahmed	NULL
5	YourFriendFirst YourFriendLast	NULL